
Topic 08

Project Evaluation

Project Evaluation is the process of assessing the performance and outcomes of a software project. The goal is to measure the success of a project, identify any issues, and provide recommendations for improvement. Project evaluation can be done at various stages of the project lifecycle and helps in understanding how effectively the project has been managed and implemented.

1. Techniques for Evaluation

Various techniques can be used to evaluate software projects, and they may differ based on the phase of the project and the metrics being considered. Below are the common techniques used:

a. Performance Reviews

- **Definition:** Performance reviews involve assessing the team's performance, both individually and collectively, in terms of meeting the project goals.
- **Example:** A project manager conducts weekly meetings to review how the team is performing in relation to the schedule, budget, and quality goals.

b. Cost-Benefit Analysis

- **Definition:** This technique involves evaluating whether the benefits of the project outweigh the costs. It's used to decide whether a project should continue or if adjustments are needed.
- **Example:** After a few months of development, the project manager compares the total cost of the project so far with the expected future benefits of the software.

c. Risk Analysis

- **Definition:** Risk analysis evaluates potential risks and how well the project is prepared to mitigate those risks.
- **Example:** A software development team conducts risk analysis to identify potential issues such as a team member's unavailability or software bugs that could delay the project.

d. Benchmarking

- **Definition:** Benchmarking involves comparing the project's progress or performance against established standards or similar projects.
- **Example:** The development team measures the software's performance against industry standards for response time or processing speed.

e. Surveys and Questionnaires

- **Definition:** Surveys and questionnaires are used to collect feedback from stakeholders about various aspects of the project.
- **Example:** A project manager sends out surveys to the client and end-users to evaluate their satisfaction with the software's functionality, usability, and overall performance.

f. Earned Value Management (EVM)

- **Definition:** EVM is a technique used to measure the project's performance by comparing the planned progress with the actual progress.
- **Example:** In EVM, if the project was planned to complete 50% of the work by a specific date, but only 40% is completed, the project manager can identify potential delays early.

2. Criteria for Evaluation

The following are some key criteria to evaluate the success of a software project:

a. Time (Schedule)

- **Definition:** The project should be evaluated based on whether it was completed within the agreed timeline.
- **Example:** If the project was planned to complete in 12 months but took 15 months, it would be evaluated negatively in terms of schedule.

b. Cost (Budget)

- **Definition:** Evaluating whether the project was completed within the allocated budget.
- **Example:** If the project was planned with a budget of \$100,000, but the final cost was \$120,000, this would be considered a negative outcome in terms of cost.

c. Quality

- **Definition:** The quality of the software product is a key evaluation criterion. It includes factors such as code quality, functionality, and usability.
- **Example:** A project is evaluated negatively if the software has a high number of bugs or doesn't meet user expectations for usability.

d. Scope

- **Definition:** The evaluation of how well the project adhered to the agreed-upon scope.
- **Example:** If the software delivered additional features that were not part of the original plan without corresponding adjustments in cost or time, it may be seen as a scope creep issue.

e. Stakeholder Satisfaction

- **Definition:** Evaluating how well the project met the expectations and needs of the stakeholders.

- **Example:** If the client is not satisfied with the software, even if it meets the budget and schedule, it is considered a failure in terms of stakeholder satisfaction.

f. Risk Management

- **Definition:** Evaluating how effectively risks were identified, assessed, and mitigated throughout the project.
 - **Example:** A project manager might evaluate whether identified risks, like a delay in a critical dependency, were properly managed without causing a major disruption.
-

3. Evaluating Project Success

a. Project Closure

- **Definition:** Evaluation at project closure involves reviewing the final product, budget, and timeline, and ensuring that all deliverables have been completed.
- **Example:** At the end of the project, the manager assesses the project's deliverables and confirms whether everything agreed upon in the contract is fulfilled.

b. Customer Feedback

- **Definition:** Gathering customer feedback through formal mechanisms like surveys, focus groups, or informal methods to evaluate the project's success from the customer's perspective.
- **Example:** After the release of the software, customer feedback surveys show that 85% of users are satisfied, which is considered a positive evaluation of the project's success.

c. Post-Mortem Analysis

- **Definition:** A post-mortem analysis is a review conducted after project completion to identify what went well and what could be improved.
 - **Example:** A project team conducts a post-mortem meeting to discuss delays due to miscommunication, but also highlights the success in quality testing and debugging.
-

Scenario-Based Questions

1. Scenario: A software project was initially planned for 8 months, with a budget of \$100,000. After 10 months, the project was completed, and the final cost was \$110,000. Additionally, the client is not satisfied with the user interface, and a few critical features were missing.

Question: How would you evaluate this project based on the criteria of time, cost, and quality?

Solution:

- **Time:** The project was delayed by 2 months. This would be a negative evaluation in terms of time.

- **Cost:** The project exceeded the budget by \$10,000, which is also considered a negative evaluation in terms of cost.
 - **Quality:** The client is dissatisfied with the user interface and missing features, indicating that the quality was not up to expectations, which is a negative evaluation.
-

2. Scenario: A software team conducted a risk analysis at the start of the project, identifying potential risks such as unavailability of key team members. They prepared a risk mitigation plan and successfully implemented it throughout the project, avoiding any major disruptions.

Question: How would you evaluate the project in terms of risk management?

Solution: The project can be evaluated positively in terms of **risk management** since the team effectively identified and mitigated risks, preventing major disruptions and ensuring smooth progress.

3. Scenario: A project was completed on time and within the budget, but many of the stakeholders feel that the final product does not meet their expectations in terms of functionality and ease of use.

Question: How would you evaluate the project in terms of stakeholder satisfaction?

Solution: Despite meeting the time and cost goals, the project would be evaluated negatively in terms of **stakeholder satisfaction** because the final product does not meet the expectations and needs of the stakeholders.

PUACP